

BIRD ISLAND, South Georgia and the South Sandwich Islands

WMO: 889000

Lat: 54.0086S

Lon: 38.0511W

Elev: 7

StdP: 14.69

Time Zone: -2.00 (W02)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.4	22.7	12.4	10.4	21.7	14.7	11.6	23.7	33.5	34.0	31.0	30.4	14.9	220	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.5	44.9	42.8	43.6	41.8	42.6	41.0	43.4	44.3	42.3	43.2	41.4	42.3	12.0	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
42.6	40.2	43.8	41.5	38.6	42.8	40.5	37.1	41.9	17.0	44.5	16.4	43.5	16.0	42.5	47.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.2	25.6	23.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	16.5	46.0	2.4	1.2	14.8	46.8	13.4	47.5	12.0	48.2	10.2	49.1	(4)
			WB													(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.6	38.4	38.7	38.2	36.0	33.5	32.2	30.6	30.6	31.9	33.5	34.8	36.9
	DBStd	4.14	2.28	2.44	2.45	3.08	3.82	3.19	3.94	3.73	3.07	2.26	2.53	2.15
	HDD50	5629	360	317	366	419	511	535	600	602	544	513	457	405
	HDD65	11104	825	737	831	869	976	985	1065	1067	994	978	907	870
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	11.4	10.5	11.2	11.6	12.0	12.3	11.6	11.9	12.1	12.0	11.2	10.3	10.5
Precipitation	PrecAvg	55.3	4.2	4.7	5.9	5.5	5.4	4.6	4.6	4.6	4.7	4.3	3.7	3.2
	PrecMax	64.9	6.7	7.3	8.7	9.5	8.0	6.5	7.3	7.3	10.0	6.8	5.3	5.0
	PrecMin	44.1	2.5	2.6	4.6	3.7	3.5	3.0	2.8	3.0	1.4	2.9	1.7	1.8
	PrecStd	6.4	1.1	1.1	1.0	1.1	0.9	0.9	1.0	1.0	1.4	0.9	0.7	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.4	47.5	46.3	44.3	42.2	39.5	39.3	38.7	40.2	40.3	42.4	44.3
		MCWB	44.2	43.9	44.3	41.8	40.5	38.1	37.8	37.3	37.7	37.9	39.6	41.7
	2%	DB	44.0	44.8	44.2	42.3	41.0	38.3	38.0	37.2	37.8	38.7	40.8	42.3
		MCWB	42.3	42.7	42.6	40.8	39.8	37.1	36.7	35.8	36.1	37.0	39.1	40.4
	5%	DB	42.7	43.4	43.0	41.4	39.9	37.3	36.9	36.2	36.7	37.8	39.5	41.3
		MCWB	41.0	41.8	41.5	40.1	38.7	36.3	35.7	34.8	35.3	36.3	38.1	39.6
	10%	DB	41.7	42.1	42.0	40.3	38.8	36.4	35.9	35.4	35.7	36.9	38.4	40.3
		MCWB	40.1	40.7	40.6	39.0	37.6	35.1	34.7	34.1	34.5	35.6	37.1	38.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.6	44.5	45.2	42.4	41.4	38.7	38.6	37.9	38.4	38.7	41.1	42.5
		MCDB	45.7	45.7	46.0	42.9	41.7	39.0	39.1	38.5	39.6	39.4	41.9	43.6
	2%	WB	42.9	43.3	43.2	41.3	40.3	37.6	37.4	36.5	36.7	37.6	39.5	41.0
		MCDB	43.7	44.3	44.0	42.0	40.9	38.0	38.0	37.2	37.4	38.4	40.4	41.9
	5%	WB	41.4	42.2	42.1	40.5	39.3	36.6	36.1	35.5	35.8	36.8	38.5	40.0
		MCDB	42.3	43.2	43.0	41.3	40.0	37.2	36.8	36.1	36.6	37.6	39.4	40.9
	10%	WB	40.4	41.0	41.0	39.3	38.0	35.6	35.0	34.4	34.9	36.0	37.5	39.2
		MCDB	41.4	42.0	42.0	40.2	38.8	36.3	35.8	35.3	35.7	36.9	38.5	40.1
Mean Daily Temperature Range	5% DB	MDBR	4.3	4.5	4.6	4.6	4.7	4.5	4.8	4.9	4.5	4.1	4.1	4.3
		MCDBR	5.3	6.3	5.7	6.1	5.1	5.3	5.5	4.7	5.6	5.3	5.4	5.5
		MCWBR	4.8	5.2	5.5	6.2	5.3	5.2	5.6	4.7	5.3	4.8	4.7	4.8
	5% WB	MCDBR	5.0	6.0	5.5	6.0	5.1	5.2	5.2	4.5	5.6	5.2	5.3	5.4
		MCWBR	4.9	5.4	5.6	6.3	5.3	5.2	5.5	4.7	5.5	4.7	4.7	4.8
Clear-Sky Solar Irradiance	taub	0.342	0.339	0.330	0.315	0.293	0.280	0.291	0.310	0.323	0.330	0.327	0.333	
	taud	2.479	2.491	2.508	2.467	2.359	2.273	2.307	2.376	2.417	2.441	2.489	2.489	
	Ebn at Noon	304	292	274	242	203	178	197	238	271	292	307	310	
	Edh at Noon	34	31	26	21	16	14	17	23	28	32	33	34	
All-Sky Solar Radiation	RadAvg	1431	1164	743	420	206	130	162	325	632	1016	1413	1617	
	RadStd	99	82	50	35	17	9	13	22	51	56	100	104	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page